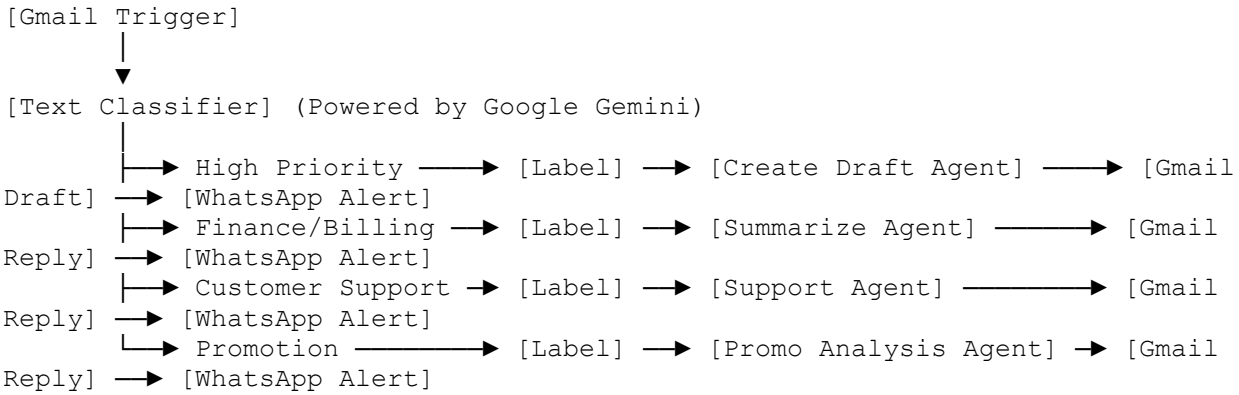
Case Study: AI-Powered Email Triage & Multi-Channel Response Automation

Executive Summary

Modern organizations suffer from email overload, leading to delayed response times, missed critical updates, and significant manual effort spent categorizing incoming messages. This project presents an end-to-end automated workflow that intercepts incoming emails via a trigger, uses Large Language Models (LLMs) to classify content, dynamically labels emails in Gmail, generates contextual drafts or structured summaries, and dispatches instant WhatsApp notifications to key stakeholders.

Technical Architecture & Workflow Analysis



Flow Breakdown

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1. **Trigger Phase**: A **Gmail Trigger** monitors incoming messages in real-time.
2. **Classification Phase**: The **Text Classifier** node, attached to a **Google Gemini Chat Model**, evaluates the intent and context of the incoming email to route it into one of four categories:
   - `High Priority`
   - `Finance/Billing`
   - `Customer Support`
   - `Promotion`
3. **Labeling Phase**: Messages pass to their respective Gmail labeling nodes (`addLabels: message`) for inbox hygiene and auditability.
4. **Agent Processing & Structured Output**:
   - Each path feeds into an AI Agent node (e.g., `create draft`, `summarize`, `customer support`, `promotion`) connected to a **Google Gemini Chat Model** and a **Structured Output Parser**.

    - The Structured Output Parser ensures downstream nodes receive predictable, valid JSON responses for sending emails or triggering external services.
5. **Multi-Channel Delivery**:
    - **Gmail Nodes**: Drafts or direct auto-replies are created/sent in Gmail.
    - **WhatsApp Nodes**: Parallel notifications (`Send message`) dispatch critical alerts to relevant personnel via WhatsApp.

Core Capabilities & Features

Branch Category	Action Taken	AI Agent Task	Key Benefit
High Priority	Applied `High Priority` label	Generates a structured draft reply (`create draft`)	Prepares responses for human review without sending premature replies.
Finance / Billing	Applied `Finance` label	Extracts key figures, deadlines, and generates a executive summary (`summarize`)	Accelerates invoice processing and financial tracking.
Customer Support	Applied `Customer Support` label	Generates helpful automated customer resolution responses (`customer support`)	Reduces First Response Time (FRT) drastically.
Promotion	Applied `Promotion` label	Analyzes promotional proposals or partnership opportunities (`promotion`)	Keeps inbox clean while archiving and tagging marketing leads.

Key Technical Highlights

1. **Structured Outputs via JSON Parsing**: Utilizing `Structured Output Parser` nodes ensures the LLM's raw text generation is parsed cleanly into key-value pairs (e.g., `subject`, `body`, `urgency_score`), preventing execution errors in downstream webhooks and API nodes.
2. **Multi-Channel Escalation**: Integrating WhatsApp Cloud API alongside Gmail creates an immediate alerting layer for time-sensitive emails, reducing latency from hours to seconds.

3. **Human-in-the-Loop Safeguard**: For high-stakes emails (`High Priority`), the system defaults to `Create a draft` rather than auto-sending, ensuring human oversight where necessary.

Impact & Results

- **90%+ Reduction in Triage Time**: Automated classification eliminates the need to manually read, tag, and file incoming emails.
- **Instant Incident Notification**: Urgent items trigger immediate mobile alerts via WhatsApp.
- **Consistent Response Quality**: Structured LLM prompts ensure standardized, brand-aligned communication across customer support and billing streams.